MODULE *MCtrace*

SCRIPTED reachability witness: drive the REAL *ccfraft* actions, in a fixed order, from the real *Init* to a state that violates *LeaderCompletenessInv*. A *step counter enables exactly one action per step, so every state is reached by genuine ccfraft transitions(messages are produced by the spec, not fabricated).*

Story(3 *nodes*, *quorum* = 2, *StartTerm* = 2; *start leader assumed* $n1$) :
- $- n1@2$ *appends a client entry* $e2$ *and signs it*($s2$), *but does NOT replicate it.*
- $- n2$ *times out*, *wins term* 3($n3$ *still only has the start log*, *so the up − to − date check lets* $n2$ *win*), *then appends/sign its own* $e3/s3 \rightarrow n2$ *DIVERGES at index* 5.
- $- n1$(*stepped down to Follower*@3 *by* $n2$ *s vote request*) times out, wins term 4 with $n3$; *BecomeLeader* truncates $n1$ to its committable prefix, which KEEPS $e2/s2$.
- $n1$ appends $e4/s4$, replicates *idx* 5 . . 8 to $n3$ (one *NACK* backup, then fills).
- *AdvanceCommitIndex*($n1$) commits the term-4 signature (*idx* 8), INDIRECTLY committing the term-2 entry $e2$ (*idx* 5). $n2$ is still *Leader*@3 with a term-3 entry at *idx* 5.

  $\Rightarrow$ *LeaderCompletenessInv* breaks.

EXTENDS *ccfraft*, *TLC*

CONSTANTS *NodeOne*, *NodeTwo*, *NodeThree*

$ToServers \triangleq \{NodeOne, NodeTwo, NodeThree\}$
$Configurations \triangleq \langle ToServers \rangle$

VARIABLE *step*
$varsS \triangleq \langle vars, step \rangle$

$$
\begin{aligned}
TInit \triangleq\ & \\
& \land step = 0 \\
& \land InitMessagesVars \\
& \land InitCandidateVars \\
& \land InitLeaderVars \\
& \land preVoteStatus = [i \in Servers \mapsto \{PreVoteDisabled\}] \\
& \land InitLogConfigServerVars(Configurations[1], JoinedLog) \quad \text{multi-node start } (JoinedLog)
\end{aligned}
$$

Each entry: < *stepGuard*, Action > . Action is a REAL *ccfraft* action with pinned params.

$Move(k, A) \triangleq step = k \land A \land step' = k + 1$

$$
\begin{aligned}
TNext \triangleq\ & \\
& \lor Move(0, ClientRequest(NodeOne)) && e2 \text{ @ } idx5\ (term2) \\
& \lor Move(1, SignCommittableMessages(NodeOne)) && s2 \text{ @ } idx6\ (term2) \\
& \lor Move(2, Timeout(NodeTwo)) && n2 \rightarrow \text{ Candidate@3} \\
& \lor Move(3, RequestVote(NodeTwo, NodeThree)) && \\
& \lor Move(4, RequestVote(NodeTwo, NodeOne)) && \\
& \lor Move(5, Receive(NodeThree, NodeTwo)) && n3\ UpdateTerm \rightarrow 3 \\
& \lor Move(6, Receive(NodeThree, NodeTwo)) && n3 \text{ grants vote} \\
& \lor Move(7, Receive(NodeOne, NodeTwo)) && n1\ UpdateTerm \rightarrow 3 \text{ (steps down)} \\
& \lor Move(8, Receive(NodeOne, NodeTwo)) && n1 \text{ rejects (more up-to-date)}
\end{aligned}
$$

$\lor\ Move(9,\ Receive(NodeTwo,\ NodeThree))$	$n2$ tallies $n3$'s grant
$\lor\ Move(10,\ Receive(NodeTwo,\ NodeOne))$	$n2$ drains $n1$'s reject
$\lor\ Move(11,\ BecomeLeader(NodeTwo))$	$n2$ *Leader*@3 (truncates to $idx4$)
$\lor\ Move(12,\ ClientRequest(NodeTwo))$	$e3$ @ $idx5$ ($term3$) – DIVERGENCE
$\lor\ Move(13,\ SignCommittableMessages(NodeTwo))$	$s3$ @ $idx6$ ($term3$)
$\lor\ Move(14,\ Timeout(NodeOne))$	$n1 \rightarrow$ Candidate@4
$\lor\ Move(15,\ RequestVote(NodeOne,\ NodeThree))$	
$\lor\ Move(16,\ Receive(NodeThree,\ NodeOne))$	$n3$ $UpdateTerm \rightarrow 4$
$\lor\ Move(17,\ Receive(NodeThree,\ NodeOne))$	$n3$ grants $n1$
$\lor\ Move(18,\ Receive(NodeOne,\ NodeThree))$	$n1$ tallies grant
$\lor\ Move(19,\ BecomeLeader(NodeOne))$	$n1$ *Leader*@4 (truncate KEEPS $e2/s2$)
$\lor\ Move(20,\ ClientRequest(NodeOne))$	$e4$ @ $idx7$ ($term4$)
$\lor\ Move(21,\ SignCommittableMessages(NodeOne))$	$s4$ @ $idx8$ ($term4$)
$\lor\ Move(22,\ AppendEntries(NodeOne,\ NodeThree))$	AE $idx7 \rightarrow n3$ rejects (gap)
$\lor\ Move(23,\ Receive(NodeThree,\ NodeOne))$	$n3$ $NACK$ ($lastLogIndex = 4$)
$\lor\ Move(24,\ Receive(NodeOne,\ NodeThree))$	$n1$ backs $sentIndex$ to 4
$\lor\ Move(25,\ AppendEntries(NodeOne,\ NodeThree))$	AE $idx5$ ($e2$)
$\lor\ Move(26,\ Receive(NodeThree,\ NodeOne))$	$n3$ appends $e2$
$\lor\ Move(27,\ Receive(NodeOne,\ NodeThree))$	$matchIndex = 5$
$\lor\ Move(28,\ AppendEntries(NodeOne,\ NodeThree))$	AE $idx6$ ($s2$)
$\lor\ Move(29,\ Receive(NodeThree,\ NodeOne))$	$n3$ appends $s2$
$\lor\ Move(30,\ Receive(NodeOne,\ NodeThree))$	$matchIndex = 6$
$\lor\ Move(31,\ AppendEntries(NodeOne,\ NodeThree))$	AE $idx7$ ($e4$)
$\lor\ Move(32,\ Receive(NodeThree,\ NodeOne))$	$n3$ appends $e4$
$\lor\ Move(33,\ Receive(NodeOne,\ NodeThree))$	$matchIndex = 7$
$\lor\ Move(34,\ AppendEntries(NodeOne,\ NodeThree))$	AE $idx8$ ($s4$)
$\lor\ Move(35,\ Receive(NodeThree,\ NodeOne))$	$n3$ appends $s4$
$\lor\ Move(36,\ Receive(NodeOne,\ NodeThree))$	$matchIndex = 8$
$\lor\ Move(37,\ AdvanceCommitIndex(NodeOne))$	commit $idx8 \rightarrow$ indirectly $idx5 \Rightarrow VIOLATION$

$$TraceSpec \triangleq TInit \land \Box[TNext]_{varsS}$$